

ARTIFICIAL INTELLIGENCE IN HEALTH COMPETITION SPECIFICATIONS

2026



VERSIONS

Version	Date	Description
V1.0	09.01.2026	Initial Release
V1.1	20.02.2026	Competition Schedule
V1.2.	14.03.2026	Competition Schedule
V1.3.	14.04.2026	Competition Schedule

CONTENTS

1. DEFINITIONS AND ABBREVIATIONS	3
2. PURPOSE OF THE COMPETITION	3
3. SCOPE OF THE COMPETITION	4
3.1. High School Level	4
3.1.1. Dataset to be Used.....	5
3.1.2. Competition Stages.....	5
3.2. University and Above Level.....	5
4. CONDITIONS OF PARTICIPATION IN THE COMPETITION.....	7
5. TEAM FORMATION	8
6. COMPETITION CALENDAR.....	9
7. SCORING AND EVALUATION	10
7.1. Evaluation of the Competition	10
7.1.1. Submission and Evaluation of the Project Presentation Report.....	10
7.1.2. Submission and Evaluation of the Project Detail Report	10
7.2. High School Level Final Evaluation.....	11
7.3. University and Above Level Final Evaluation.....	11
7.4. General Evaluation and Scoring.....	11
7.5. Scoring of the Competition.....	11
7.6. Question & Answer Meetings with Teams	12
7.7. Datasets to be Shared with Competitors	12
8. AWARDS	12
8.1. Best Presentation Awards	13
8.2. Issues Regarding Awarding.....	13
9. GENERAL RULES.....	13
10. ETHICAL RULES.....	13
11. COMMUNICATION.....	14
12. STATEMENT OF LIABILITY	14
12.1. Use/Processing of Personal Data.....	15

1. DEFINITIONS AND ABBREVIATIONS

As referred to in this specification;

Bank Notification Form: The printed form in which the person to whom the payment will be made declares their bank details and necessary personal information.

Advisory and Assessment Committee: The committee consisting of experts in their fields, tasked with duties such as subject determination, dataset creation and organization, report evaluation, specification revision, report template creation, answering technical questions, and creating the competitor information kit in the TEKNOFEST Artificial Intelligence in Health Competition.

CMS (KYS): TEKNOFEST Corporate Management System.

High School Level: Competitions participated in by individuals currently studying in high school or whose last graduated school is high school and who are not enrolled in any other educational institution.

Confidentiality Agreement: The "Corporate Confidentiality Undertaking" signed by the competitors in order to use the anonymized information/documents/data shared by the Ministry of Health of the Republic of Türkiye / TUSEB for the purpose of training and/or testing the competitors' models within the scope of the Artificial Intelligence in Health Competition.

Stakeholder: The Ministry of Health of the Republic of Türkiye, the Health Institutes of Türkiye (TUSEB), and the T3 Foundation.

System: The arrangement containing all kinds of hardware and software necessary for the competitors' solutions and proposals to be in working order.

Team: The group consisting of at least 2 (two) or at most 5 (five) persons.

Team Captain: The team member authorized to represent the team in the competitions.

Team Advisor: The teacher/instructor/academician, limited to a maximum of 1 (one) per team, which is mandatory for the High School level and optional for the University and Above levels.

TEKNOFEST: Aviation, Space and Technology Festival.

TEKNOFEST Competitions Committee: The committee formed by the relevant responsible members from the stakeholder institutions.

TUSEB: Health Institutes of Türkiye (Türkiye Sağlık Enstitüleri Başkanlığı).

T3 Foundation: Turkish Technology Team Foundation (Türkiye Teknoloji Takımı Vakfı).

Competition Process: Denotes the period elapsing between the date when competition applications start being received and the date when the final results are announced.

2. PURPOSE OF THE COMPETITION

In recent years, artificial intelligence has assumed an increasingly visible and effective role in the field of health. Artificial intelligence-based solutions, utilized in many stages from diagnosis to treatment and from risk prediction to clinical decision support processes, increase both the accuracy and efficiency of healthcare services. Based on this approach, the TEKNOFEST Artificial Intelligence in Health Competition encourages the development of innovative, applicable, and high value-added artificial intelligence solutions for the health sector. In this year's competition, unlike the field of radiology which was predominantly addressed in previous years, Genetics and Cardiology have been determined as priority focus areas due to their high data volume and potential for strong clinical impact. While solutions regarding early diagnosis of diseases and

personalized treatment approaches stand out in the field of genetics; artificial intelligence applications supporting early detection of cardiovascular diseases, risk assessment, and clinical decision processes come to the forefront in the field of cardiology.

High school level teams shall develop models capable of recognizing basic clinical conditions such as cardiac rhythm disorders and conduction abnormalities by processing Electrocardiography (ECG) data using artificial intelligence methods. The selection of this field will allow competitors to work on detailed ECG findings representing critical clinical decision points by using signal processing and machine learning architectures.

University and above level teams shall develop prediction models regarding the status of variants whose clinical condition is unknown as "Pathogenic (disease-causing)" or "Benign" by using the biological and computational properties of a small number of known variants. It is aimed to contribute to the literature by making predictions on unknown variants starting from the biological properties of the small number of known variants with the models to be developed by the competitors.

Within the scope of TEKNOFEST 2026, teams shall develop their models using open data sources along with the datasets provided regarding the determined problems. The developed models shall be tested with new data during the competition, and an external validation process shall be carried out. External validation, which is a critical stage in the transition to clinical applications, is important for demonstrating how effective the models developed by the teams using various data are on new data.

Consequently, this year's competition setup is planned to serve directly to ensure that artificial intelligence solutions developed in data-intensive fields such as genetics and cardiology reach a maturity capable of responding to real health system needs by emerging from the laboratory environment.

3. SCOPE OF THE COMPETITION

3.1. High School Level

In the first stage of the competition, participants are expected to develop a model that classifies four main cardiac conditions using 12-lead ECG data.

Participants are expected to develop a model capable of distinguishing the following main cardiac conditions from ECG data:

1. Arrhythmias
2. Conduction Blocks
3. Normal ECG

Output Format: A single super-class prediction shall be made for each record. (Multi-class)

In the second stage, participants shall develop a "fine-grained" classification model that distinguishes more specific cardiac diagnoses under the two super-classes presented in the first stage.

Sub-group examples are given below:

- Arrhythmias → AFIB, AFL, AT, SVT, APB, VPB, etc.
- Conduction Blocks → LBBB, RBBB, AV block types, etc.

Note: The definitive list of sub-groups will be announced after the Project Presentation Report (PPR).

3.1.1. Dataset to be Used

In the competition, the ECG Arrhythmia Dataset (1.0.0) shared by PhysioNet will be presented as sample data*. This dataset contains 12-lead ECG recordings, a 500 Hz sampling frequency, signals from approximately forty-five thousand individuals, and diagnostic labels compliant with SNOMED-CT standards. Data is provided in mat and hea formats, and competitors will be able to access signal and metadata from these files. Competitors will also be able to use different open-access datasets and/or datasets they create themselves for model training.

*(<https://physionet.org/content/ecg-arrhythmia/1.0.0/>)

3.1.2. Competition Stages

First Stage: Cardiac Classification

In this stage, competitors shall develop a model that matches each ECG record with one of the super-classes. As these classes require learning the fundamental patterns of the ECG, this offers participants a comprehensive field of study regarding signal processing, feature extraction, time-frequency analysis, and machine learning architectures.

The official evaluation metric of the first stage shall be the macro F1-score. The reason for using Macro F1 is that the data distribution among super-classes may be imbalanced, and therefore metrics such as accuracy may be misleading. During the evaluation process, confusion matrix analysis, learning curves, and class-based errors will also be examined.

Second Stage: Classification of Sub-Groups

In the second stage, participants shall develop a classification system distinguishing sub-diagnostic groups belonging to one of the two super-classes. Since this stage involves detailed ECG findings representing more critical decision points from a clinical perspective, it strengthens the scientific aspect of the competition. The definitive list of sub-classes shall be determined by the Advisory Board based on clinical significance, data distribution, and expert opinions, and shall be announced to the competitors prior to the second stage.

3.2. University and Above Level

In the classification of variants contained in the datasets as Pathogenic or Benign, the American College of Medical Genetics and Genomics (ACMG) guidelines and criteria have been taken as reference. In the classification process, existing ACMG-compliant tags found in source databases have been accepted as reference and included in the dataset. These tags shall be used as 'Ground Truth' within the scope of the competition. The datasets to be provided to the competitors have been prepared according to the following standards:

Pathogenic Class: Missense variants with "Expert Panel" and reliable "Practice Guideline" review status, and 3 and 4-star reliability levels, were selected from ClinVar and ClinGen databases. Variants defined as 'Pathogenic' and 'Likely Pathogenic' in the datasets have been merged under a single Pathogenic class (2909 variants).

Benign Class: In order to address class imbalance, in addition to ClinVar (1381 variants) data, frequently observed healthy population variants of the

genes in the ClinVar data from the gnomAD database will be added (~1500 variants). Variants defined as 'Benign' and 'Likely Benign' in the datasets have been merged under a single Benign class.

Competitors shall be presented with datasets under four main headings. In the creation of the datasets, data balance has been observed to prevent bias in the training processes of artificial intelligence models. In this context, a balanced structure has been created by keeping the number of Benign variants in each dataset close to the number of Pathogenic variants.

The variant numbers contained in the training datasets to be provided are as follows:

1. **General Dataset:** 1500 Pathogenic variants, 1500 Benign variants.
2. **Hereditary Cancer Panel:** 200 Pathogenic variants, 200 Benign variants.
3. **Phenylketonuria (PAH) Gene Panel:** 200 Pathogenic variants, 200 Benign variants.
4. **Cystic Fibrosis (CFTR) Gene Panel:** 70 Pathogenic variants, 70 Benign variants.

The variant numbers contained in the test datasets to be provided are as follows:

1. **General Dataset:** 1000 Pathogenic variants, 1000 Benign variants.
2. **Hereditary Cancer Panel:** 100 Pathogenic variants, 100 Benign variants.
3. **Phenylketonuria (PAH) Gene Panel:** 100 Pathogenic variants, 100 Benign variants.
4. **Cystic Fibrosis (CFTR) Gene Panel:** 30 Pathogenic variants, 30 Benign variants.

The total data pool has been divided into two for the healthy execution of model development processes:

Training Set: The labelled training set to be given to competitors who pass the Project Presentation Report stage to train their models.

Test Set: This set, separated to evaluate the final performance of the developed models, will be shared with class labels hidden (unlabelled) during the competition.

Genomic address (chromosome and position) information of the variants in the competition dataset has been strictly hidden to prevent participants from finding the label directly by referring to external data sources. The purpose of this restriction is to ensure that competitors make their pathogenicity predictions solely based on the variant profiles provided by the competition committee without resorting to external data sources and to prevent the use of ready-made label information obtainable from public databases.

In order for the models to be trained with high success, comprehensive and multi-dimensional variant profiles enriched with bioinformatics tools will be presented instead of genomic address (chromosome and position) information. The feature set provided for each variant covers the following details; feature column names will not be given when the data is shared:

- **Sequence and Change Information:** Reference and alternative nucleotide information at the point where the variant occurs, codon change, and the amino acid conversion caused by this change (e.g., conversion from Alanine to Valine).
- **Local Sequence and Environmental Context Information:** In order for the model to learn the structural characteristics and local motifs of the region where the variant is located; the 5 nucleotides preceding and succeeding the variant point (genomic neighbourhood) and the 5 amino acids preceding and succeeding the relevant amino acid at the protein level (proteomic neighbourhood) information shall be provided.
- **Biochemical and Structural Effects:** Potential effects of the occurring amino acid change on the physicochemical properties (hydrophobicity, polarity, molecular weight change, etc.) and 3D structure of the protein.
- **Evolutionary Conservation:** Genomic similarity of the gene region where the variant is located across different living species (phylogenetic diversity) and human populations; scores indicating how well this region has been conserved during the evolutionary process.
- **Population Data:** Frequency of occurrence of the variant in different human populations (Minor Allele Frequency, etc.).
- **In Silico Risk Scores:** Computational risk scores regarding the probability of the variant being harmful, calculated by different algorithms.

4. CONDITIONS OF PARTICIPATION IN THE COMPETITION

- High school students and graduates studying in Türkiye and abroad may apply to the High School Level; university students (including Undergraduate, Associate Degree, Master's, PhD, and Open Education) and graduates may apply to the University and Above Level.
- Approved student certificates must be submitted to www.t3kys.com together with the Project Presentation Report.
- For the High School Level: The literature review for the First Stage, methods used/to be used, open datasets, and preliminary results obtained shall be clearly presented in the Project Presentation Report. In the Project Detail Report, the literature regarding the Second Stage, preferred methods, datasets used, and results obtained shall be discussed in detail. At the end of each stage, prediction-generating codes, used datasets, and result files may be requested if necessary, along with a technical report containing model training, pre-processing processes, and evaluation methodologies from the competitors. Codes are expected to be executable, reproducible, and clearly documented.
- For the University and Above Level: The definition of the general problem, literature review regarding this problem, and details of the proposed solution method shall be clearly presented in the Project Presentation Report. In the Project Detail Report, the architecture of the developed model, training processes, and internal test (validation) results shall be presented in detail.
- Report templates will be shared via the TEKNOFEST website.
- Members of the same team may apply to other competitions organized within the scope of TEKNOFEST with different projects.

- Competitors shall be able to access the data to be provided by the stakeholders in the competition and participate in the competition only if they submit the signed "Confidentiality Agreement".
- The competitor shall be able to participate in the competition by reading and approving all explanations and participation conditions in this specification before applying. In case the competitor is under 18 (eighteen) years of age, submission of a document showing that this specification has been approved by their parent/guardian is mandatory.
- Applications are made online via the www.t3kys.com application system until 20.02.2026.

5. TEAM FORMATION

- During the application, the Team Authority registers via the system, registers the team members accurately and completely into the system, and sends invitations to the e-mails of the advisor (if any) and the members. The invited member logs into the application system, accepts the incoming invitation from the "My Team Information" section, and the registration is completed. Otherwise, the registration shall not be deemed completed.
- Competitors who have completed the team formation process are required to apply to the competition suitable for their project.
- All processes required within the scope of the competition (Application, Report Receipt, Report Results, Objection Processes, Member Addition/Removal Procedures, etc.) are carried out via the CMS. Teams are required to follow their processes via the CMS.
- Member addition/removal procedures shall be performed until the Project Detail Report submission deadline.
- During the competition process, the procedures of applying, uploading reports, and filling out forms via the CMS are under the responsibility of the Team Authority, and the competition processes are managed through this Team Authority.
- Transportation and accommodation support to be provided to the finalist teams is limited. The number of persons to be supported is 3 (three) per team (including the advisor), and the TEKNOFEST Competitions Committee reserves the right to make changes.
- A member of a team shall not participate as a member of another team in the same competition.
- In the competition, teams may consist of a minimum of 2 (two) and a maximum of 5 (five) persons. (The advisor is not included in this number.)
- Teams can be formed from a single school, or they can be formed as mixed teams by bringing together one or more secondary education/higher education students.
- The TEKNOFEST Competitions Committee has the authority to limit the number of members to be present in the festival area. In case of a limitation, information will be provided by the committee.
- It is mandatory to have an advisor in High School Level teams. It is not mandatory to have an advisor in applications to be made for the University and Above Level

- The primary duty of the advisor is to assist students in planning their own education and training; to guide them in academic, social, and cultural matters; and to help prepare a suitable environment for the development of the student's personality as a whole in mental, social, and emotional aspects. The specific role of the advisor in the team is to guide the team members to produce solutions to problems by providing the academic support needed in the project. The project idea is required to be created by the team members. Advisors are only requested to provide guidance at the idea stage. If it is determined that the project owner is the advisor, the application shall be deemed invalid.
- The advisor shall not be added as a team member.
- It is mandatory for the person who will serve as an advisor to submit a wet-signed/electronically signed scanned document proving that they will fulfil their consultancy duties and that they are a teacher/instructor/academician or an engineer/expert etc. continuing their career in the relevant field, and the assignment letter to be obtained from the institution they work for, to the TEKNOFEST Competitions Committee (www.t3kys.com) until the Project Detail Report submission date.
- Although the model, code, and all kinds of reports produced at every stage of the competition are formed as a result of the labor of the competitor/competitors in the team and reflect the characteristics of the team members, the Advisor shall not be accepted as the author/owner of the work.
- The advisor commits to supporting the team until the final stage and being with the team during the final stage. In projects that make it to the final stage, teams are expected to be in the field with their advisors.
- In case of a change of advisor, teams may update their requests regarding this change via the CMS during the team member addition/removal procedures.

6. COMPETITION CALENDAR

DATE	DESCRIPTION
15.03.2026	Competition Application Deadline
23.03.2026	Q&A Meetings I (High School and University)
25.03.2026 (17:00) GMT (+3)	Project Presentation Report Submission Deadline (High School and University)
22.04.2026	Announcement of Teams Passing the Preliminary Elimination based on Project Presentation Report Results
05.05.2026	Data Sharing (High School and University Category)
18.05.2026	Q&A Meetings II (High School and University)
29.06.2026 (17:00) GMT (+3)	Project Detail Report Submission Deadline (High School and University)
20.07.2026	Announcement of Project Detail Report Results and Finalist Teams
August-September 2026	Competition Date
30 September-4 October	TEKNOFEST Şanlıurfa

7. SCORING AND EVALUATION

7.1. Evaluation of the Competition

Project Presentation and Project Detail Reports shall be subjected to preliminary elimination by the Advisory and Assessment Committee. The scores obtained at this stage shall only be evaluated for elimination purposes and will not affect the final score. Reports falling outside the scope of evaluation shall not be processed, and no objections shall be accepted for eliminations resulting from this situation. The Project Detail Report to be submitted by the teams passing the Project Presentation Report stage shall be evaluated by the Advisory and Assessment Committee. Those who pass the evaluation according to the Project Detail Report shall be entitled to participate in the final. Finalists shall be evaluated by the Advisory and Assessment Committee in accordance with the specified Final Evaluation, and the winners shall be announced. The Advisory and Assessment Committee reserves the right to change the performance measurement metrics used in the evaluation processes if deemed necessary.

7.1.1. Submission and Evaluation of the Project Presentation Report

Teams must upload the Project Presentation Report to the CMS by the date specified on the TEKNOFEST website and in accordance with the report format to be published. Detailed information regarding the submission of Project Presentation Reports will be shared with the teams that have completed their applications after the competition application deadline. These submitted reports will be examined by the Advisory and Assessment Committee, and the teams entitled to continue in the competition will be announced. After the Project Presentation Reports are evaluated and scored by the Assessment Committee, teams scoring at or above the threshold score will be able to proceed to the next stage. Teams that do not submit a Project Presentation Report, submit a report outside the scope, or whose report scores fall below the threshold score shall not be able to participate in the next stage. Teams falling below the PPR threshold score by a maximum of 15 points have the right to object. Teams falling below the threshold score by more than 15 points do not have the right to object. The project presentation report template will be accessible via the website.

7.1.2. Submission and Evaluation of the Project Detail Report

The Project Detail Report template will be prepared specifically for the competition and presented under the Artificial Intelligence in Health Competition on the TEKNOFEST website. Teams passing to the Project Detail Report stage must upload their reports to the CMS by the date specified on the TEKNOFEST website and in accordance with the presented report template. After the Project Detail Reports are evaluated and scored by the Advisory and Assessment Committee, teams scoring at or above the threshold score to be determined by the TEKNOFEST Competitions Committee will be able to proceed to the final stage. Teams falling below the PDR threshold score by a maximum of 15 points have the right to object. Teams falling below the threshold score by more than 15 points do not have the right to object. Teams entitled to participate in the final will be announced on the date specified in the Competition Calendar.

7.2. High School Level Final Evaluation

During the competition process to be held at the TEKNOFEST 2026 festival area, an external validation will be performed to test the model performance within clinical reality. In this direction, a new, unique, and completely anonymized ECG dataset for TEKNOFEST will be used. This dataset will be used both to test how the algorithms developed by the competitors behave in different patient groups and to evaluate their generalization ability. The external validation dataset will be used solely for research, competition evaluation, and training purposes, and all legal regulations regarding the protection of personal data will be strictly adhered to.

The evaluation metric at the high school level final stage will be the macro F1-score.

7.3. University and Above Level Final Evaluation

The success of the models will be measured by comparing the predictions on the test data with the actual labels. **The primary metric that will determine the competition ranking will be the F1 Score calculated over TP (True Positive), FP (False Positive), and FN (False Negative) values**

7.4. General Evaluation and Scoring

The contribution of the tasks given for each level (high school/university and above) to the total competition score will be 90%. The presentation made in the final stage constitutes 10% of the total score. Total scores will be formed by adding the presentation scores to the scores obtained from the tasks. Success rankings will be made considering this score. The top three teams according to the success ranking will be entitled to receive the determined awards.

7.5. Scoring of the Competition

Scoring types and percentages are specified in the table below.

SCORING TYPE	SCORING PERCENTAGE
Project Presentation Form	0
Project Detail Report	0
Final Competitions (Physical)	%90
Final Presentation Scoring	%10

Algorithm/poster/application presentations of the finalists are evaluated personally by the Assessment Committee in the event area. The Assessment Committee records the result with a report in the competition area. Each team qualifying for the finals must ensure all members have filled out the Bank Notification Form (including advisors) (present and not present in the area) and deliver it signed to the TUSEB Competitors Officer before leaving the competition area. The competition jury has the authority to ask the finalist teams to re-run their codes and find the results they declared.

TUSEB reserves the right to make changes to the evaluation metrics and evaluation rates used in the competition evaluation stage.

7.6. Question & Answer Meetings with Teams

After the competition applications are completed, Q&A Meeting-I will be held with the teams. Information regarding the meeting will be announced later.

Q&A Meeting-II will be held with the competitors who pass the project presentation stage in order to answer questions about the competition. Competitors will be informed in case of a change in the meeting date and about the meeting environment.

7.7. Datasets to be Shared with Competitors

For the High School level, no data sharing will be done at the Project Presentation Report stage. Competitors will utilize open datasets. The dataset containing sub-groups will be shared with the competitors who pass the Project Presentation Report evaluation.

For the University and above level, labelled data belonging to Pathogenic and Benign classes will be shared with the teams that pass the Project Presentation Report stage. Details of the datasets are given in Section 3.2.

8. AWARDS

Cash prizes shall be awarded to teams that pass all stages as a result of separate evaluations in three stages in the competition, become finalists in their own category, and are ranked in the final evaluation. The awards specified in the table below show the total amount to be given to the teams entitled to receive an award; no individual awarding shall be made. First, second, and third place awards shall be divided equally according to the total number of existing team members (excluding the advisor) in your team in the CMS and deposited into the bank account specified by each competitor during the application. An "Advisor Award" payment shall be made to the advisor of the ranked team within the scope of the competition.

Category	First	Second	Third
High School Level	150.000,00 ₺	120.000,00 ₺	100.000,00 ₺
University and Above Level	180.000,00 ₺	150.000,00 ₺	130.000,00 ₺
Advisor	15,000.00 ₺	12,000.00 ₺	10,000.00 ₺

(Taxes, duties, and similar expenses are included in the payments to be made, and the net value to be obtained by the competitor shall consist of the remaining amount after the said legal deductions are subtracted.

8.1. Best Presentation Awards

Finalist teams shall prepare their presentations in the presentation format determined by TEKNOFEST. These presentations shall be evaluated by the Advisory and Assessment Committee members in terms of presentation techniques and awarded with a plaque under the name "Best Presentation Award" on behalf of the team. The presentation made shall be evaluated independently of whether the Team ranks in the competition or not. The specified award is for prestige purposes and has no monetary value.

8.2. Issues Regarding Awarding

- Awards are valid for teams fulfilling the success criteria.
- Awards shall be given to the teams and shared equally among team members.
- For foreign national team members who do not have an account in a bank operating in Türkiye, they are required to designate a person resident in Türkiye as a trustee/representative.
- Payments shall be made to the competitors' own bank accounts, or depending on the situation, to their proxy, parent, guardian, or trustee for competitors who are minors, provided that it is documented in accordance with the Central Governance Expenditure Documents Regulation. The Bank Notification Form must be filled out accordingly.
- Competitors must submit all documents regarding payment to TUSEB with wet signatures in order for the payment to be made.
- For the advisor of the ranking team at the High School Level (including persons performing the same duty), a payment of at most the total amount given in the table above shall be made per competitor or team, regardless of the number [of advisors].
- Regarding other matters related to awarding, the provisions of the Health Institutes of Türkiye Award Regulation published in the Official Gazette dated 25.09.2021 and numbered 31609, and the Central Governance Expenditure Documents Regulation published in the Official Gazette dated 31.12.2005 and numbered 26040 shall be taken as the basis.

9. GENERAL RULES

[Click](#) to access the General Rules booklet valid within the scope of the competition.

10. ETHICAL RULES

[Click](#) to access the Ethical Rules booklet valid within the scope of the competition.

All genetic data used within the scope of the competition are obtained from publicly available and anonymized databases managed by international scientific consortia such as ClinVar, ClinGen, and gnomAD.

The genetic data used are data that have already been collected by primary collector institutions (NCBI, Broad Institute, etc.) in accordance with international bioethical standards (Declaration of Helsinki) and relevant patient consent processes. The operation performed within the scope of the competition is in the status of "secondary data use".

The datasets presented to the competitors do not contain any Personally Identifiable Information (PII) that could reveal the identity of the participants (name, surname, date of birth, etc.), in accordance with the Law on the Protection of Personal Data (KVKK) and international GDPR standards. Additionally, due to the competition format in the university and above category, the genomic address information of the variants has also been removed, and the risk of "re-identification" has been technically eliminated. The data controller regarding the datasets provided within the scope of the competition is the TEKNOFEST organization. Competitors are obliged to use the relevant data only within the scope determined by the organization and in the capacity of a data processor.

The use of data does not pursue a commercial purpose; it carries the purpose of purely scientific algorithm development. This situation aligns perfectly with the principle of "use of scientific progress for the benefit of humanity" of the UNESCO Universal Declaration on the Human Genome and Human Rights.

The models developed and outputs obtained within the scope of the competition cannot be used for any clinical diagnosis, treatment, or medical decision support purposes. These outputs are for research and educational purposes only.

11. COMMUNICATION

For questions regarding the competition, you may join the competition group via the Artificial Intelligence in Health Competition page on the TEKNOFEST website. It is the responsibility of the competitors to actively monitor this group, and for at least 1 (one) person from each team to become a member and follow the announcements, questions, and answers in this group. The TEKNOFEST Competitions Committee shall not be held responsible for the teams' failure to access up-to-date information resulting from the failure to monitor the specified e-mail group.

Questions regarding the organizational aspects of the competition must be submitted via the iletisim@teknofest.org e-mail address.

Submitting your questions via the correct channels mentioned above is important for providing quick responses to the questions asked.

12. STATEMENT OF LIABILITY

TEKNOFEST and Stakeholder Institutions shall in no way be held responsible for any injury or damage arising from any product submitted by the competitors or caused by the competitor. TEKNOFEST, Stakeholder Institutions, and organization officials shall not be held responsible for damages caused by competitors to third parties. TEKNOFEST and Stakeholder Institutions shall not be held responsible for ensuring that teams prepare and implement their own systems within the framework of the laws of the Republic of Türkiye.

Turkish Technology Team Foundation reserves the right to make any amendments to this specification.

12.1. Use/Processing of Personal Data

Personal data of persons applying to the competition (TR Identity Number, e-mail, phone number, IBAN, Certificate of Identity Register Copy) may be processed and transferred by TUSEB regarding the works and transactions carried out within the scope of award payments and promotional activities. Said personal data is processed based on the provisions of subparagraph (ç) of the second paragraph of Article 5 of the Law on the Protection of Personal Data No. 6698, stating "It is mandatory for the data controller to fulfill its legal obligation," and subparagraph (f) of the second paragraph of Article 5, stating "Data processing is mandatory for the legitimate interests of the data controller, provided that it does not harm the fundamental rights and freedoms of the data subject."

It is accepted that the personal data sent by you to TUSEB has been obtained in accordance with the legislation and that the necessary technical and administrative measures regarding this personal data have been taken by you. Personal data obtained in this way may be transferred to authorized public institutions and organizations, real persons, and private law legal entities in accordance with the purpose of the relationship between the parties and Article 8 of Law No. 6698. You may exercise your rights within the scope of Article 11 of Law No. 6698 regarding your personal data transmitted to TUSEB as the data controller.

TEKNOFEST and Stakeholder Institutions reserve the right to make all kinds of amendments to this specification and related documents (report templates, forms, etc.).

Applicants to the competition are deemed to have accepted all conditions in this specification.

